



Pharm*Achieve*

COVID-19

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

ACE-2 Angiotensin Converting Enzyme 2

ARDS Arterial Respiratory Distress Syndrome

BMI Body Mass Index

BPM Breaths Per Minute

CHF Congestive Heart Failure

CKD Chronic Kidney Disease

COPD Chronic Obstructive Pulmonary Disease

COVID-19 Coronavirus Disease 2019

CRP C-Reactive Protein

CVD Cardiovascular Disease

DVT Deep Vein Thrombosis

eGFR estimated Glomerular Filtration Rate

FiO₂ Fraction of Inspired Oxygen

IL Interleukin

mAb Monoclonal Antibody

MERS Middle East Respiratory Syndrome

MERS-CoV Middle East Respiratory Syndrome coronavirus

NAAT Nucleic Acid Amplification Test

NSAID Non-Steroidal Anti-inflammatory Drug

NT-proBNP N-Terminal Pro-B-Type Natriuretic Peptide

PaO₂ Arterial partial pressure of Oxygen

PCR Polymerase Chain Reaction

PE Pulmonary Embolism

PO Per Os (By Mouth)

RR Respiratory rate

SARS-CoV-2 Severe Acute Respiratory Syndrome Coronavirus 2

SIRS Systemic Inflammatory Response Syndrome

SOB Shortness Of Breath

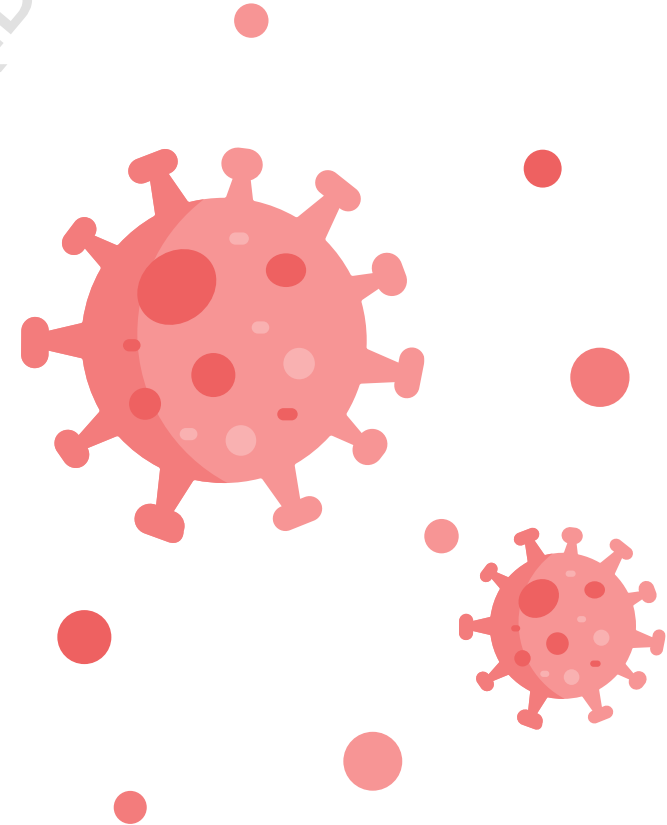
SpO₂ Oxygen Saturation

VTE Vein Thromboembolism

WHAT IS COVID-19?

FYI

- COVID-19 is an acute infectious respiratory disease caused by the SARS-CoV-2 virus which belongs to the coronavirus family
- This is the same family of viruses that caused the SARS (SARS-CoV) and MERS (MERS-CoV) epidemics of 2003 and 2012



VIROLOGY

FYI

- Enveloped positive stranded RNA virus
- Host receptor for cell entry: angiotensin-converting enzyme 2 (**ACE-2**)
- SARS-CoV-2 has undergone many mutations, creating different strains such as alpha, delta, etc.

ROUTE OF TRANSMISSION & INCUBATION PERIOD

- SARS-CoV-2 causes disease when it is **inhaled** into the lungs
- The virus has also been found to survive on **faecal matter** (improper hand hygiene), and **inanimate surfaces for up to 7 days**
- The virus can travel from a person's hand to their mouth, nose or eyes
- Incubation periods may vary between strains, ranging from 2-14 days
 - During this period, the infected patient may test positive before experiencing symptoms
- Patients may also **test positive up to 90 days after symptom resolution**

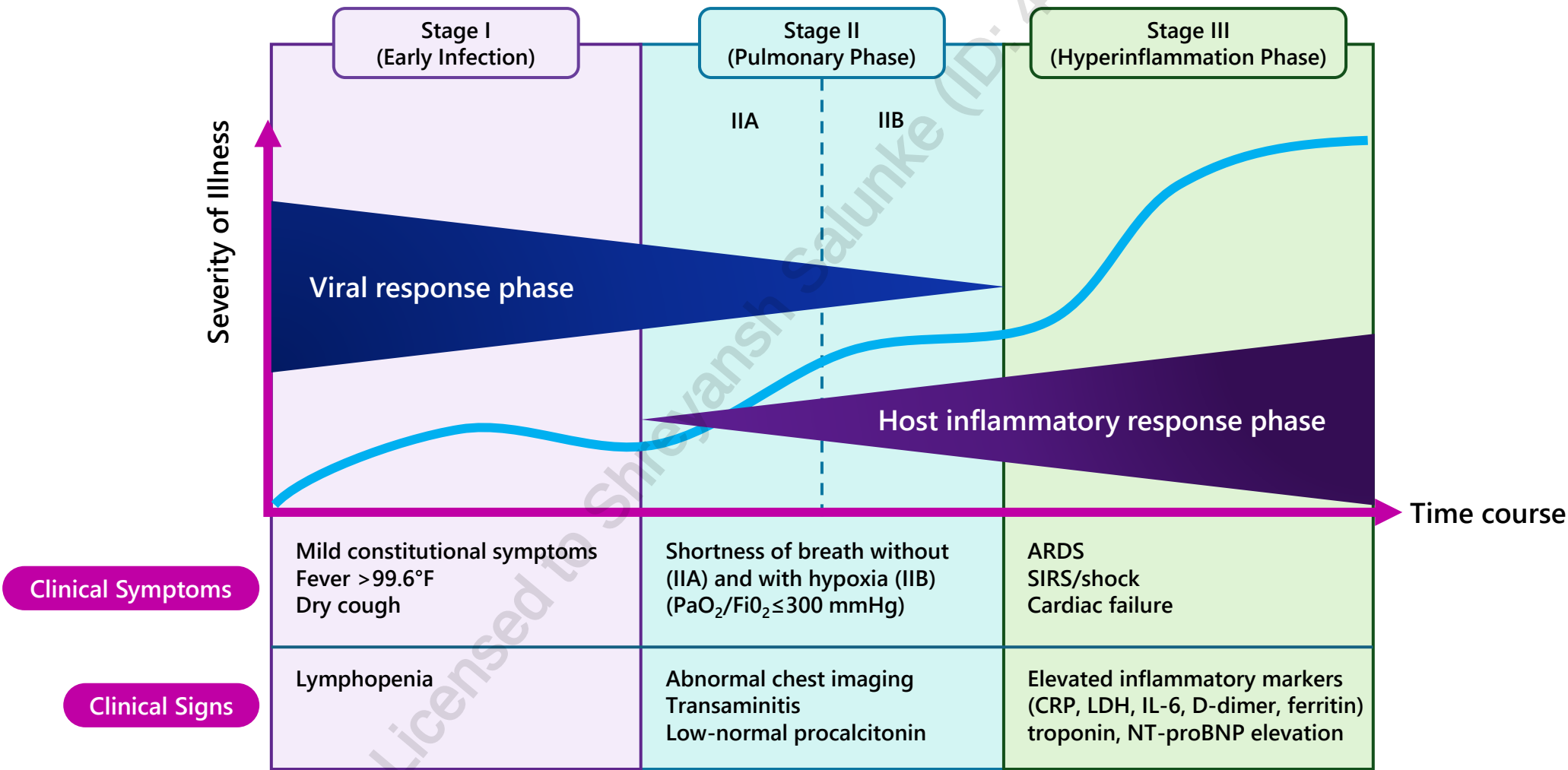
COVID-19 SEVERITY SPECTRUM

FYI

- Asymptomatic or Pre-symptomatic Infection
 - Positive test for SARS-CoV-2 without any symptoms
- **Mild** illness
 - Fever, cough, loss of taste, etc.
 - No shortness of breath or abnormal chest imaging
- **Moderate** illness
 - Signs of lower respiratory disease either by imaging or clinical assessment and $\text{SpO}_2 \geq 94\%$
- **Severe** illness
 - An $\text{SpO}_2 < 94\%$, a $\text{PaO}_2/\text{FiO}_2 < 300$ mmHg, a RR > 30 bpm, or lung infiltrates $> 50\%$
- Critical illness
 - Respiratory failure, septic shock, and/or multiple organ dysfunction

PHASES OF COVID-19 DISEASE PROGRESSION

FYI



COMPARISON OF SIGNS AND SYMPTOMS

Signs & Symptoms	Cold	COVID-19	Influenza (Flu)	Pneumonia	Strep Throat
Sore throat	✓	✓	✓		✓
Stuffy/Runny Nose	✓	✓	✓		
Cough	✓	✓	✓	✓ (dark sputum)	
Fever and/or Chills		✓	✓	✓	✓
Headache		✓	✓		✓
Fatigue		✓	✓	✓	
Muscle pain/aches		✓	✓	✓	
Vomiting and/or Diarrhea		✓	✓		✓ (Possible vomiting)
Change in or loss of taste or smell		✓			
Shortness of Breath		✓	✓	✓	

COVID-19 TESTING

- **Polymerase Chain Reaction (PCR) – Gold Standard** (nasal or nasopharyngeal swab)
 - PCR is a type of Nucleic Acid Amplification Test (NAAT): it detects one or more viral RNA genes and amplifies it
 - Highly sensitive and specific
 - Costly and results are delayed (>1 hour – days)
- **Rapid Antigen** (nasal or nasopharyngeal swab)
 - Antigen tests are a type of immunoassay that detects presence of specific viral proteins (antigens)
 - More accessible, and results are obtained within 15 minutes
 - Less sensitive than NAAT testing, may need to repeat testing to ensure negative result

RED FLAGS



- Cold, clammy, pale skin
- Confusion
- Reduced level of consciousness
- Decreased urine output
- Difficulty breathing
- Hemoptysis
- Non-blanching rash
- Shortness of breath at rest
- Worsening of respiratory symptoms (increased fatigue, blue lips/face, syncope, pain/pressure in the chest)

RISK FACTORS FOR SEVERE DISEASE

- Unvaccinated population
- Immunocompromised e.g. cancer, immunosuppressant therapies, autoimmune diseases
- Chronic medical conditions including but not limited to: CVD, CKD (eGFR <60 mL/min), diabetes, COPD, CHF, Cystic Fibrosis
- Pregnancy
- Old age (>60, risk increases with increasing age)
- Obesity (BMI ≥ 40)
- Sickle Cell Disease
- Moderate to severe liver disease

GOALS OF THERAPY



COVID-19 COMPLICATIONS

- Neurological (i.e. stroke)
- Respiratory (i.e. mechanical ventilation, ARDS)
- Cardiovascular (i.e. arrhythmias, acute cardiac injury)
- VTE (i.e. DVT, PE)
- Inflammatory (i.e. persistent fevers, elevated inflammatory markers and pro-inflammatory cytokines)
- Recovery and long-term sequelae-Long COVID (i.e. fatigue, SOB, chest pain, cognitive deficits, lack of smell/taste, anxiety, insomnia)

SYMPTOM MANAGEMENT FOR COVID-19

- Prioritize hydration and drink plenty of fluids
- Rest and eat nourishing foods to support immune system
- Cough: a spoonful of honey, dextromethorphan PRN (≥ 6 years)
- Fever/Headache/Myalgia: Acetaminophen or NSAIDs
- Prevent the spread of infection to others:
 - Wash hands frequently
 - Cover face when coughing or sneezing
 - Consider staying home or wearing a mask if going out

PREVENTION-VACCINES

- **Comirnaty® Bivalent** (previously **Pfizer**-BioNTech) Mono/Bivalent **mRNA** vaccine
 - Consists of 30 mcg, 10 mcg or 3 mcg of mRNA encoding for the spike protein of Omicron KP.2 variant), Pfizer and BioNTech Manufacturing GmbH
 - Authorized for use in those 12 years of age and older (30 mcg), children 5 to 11 years of age (10 mcg), and children 6 months of age to less than 5 years of age (3 mcg)
 - Note: Only the 30 mcg dose is available beginning in Fall 2024 (for individuals 12 years of age or older)
- **Spikevax® Bivalent** (previously **Moderna**) Mono/Bivalent **mRNA** vaccine
 - Consists of 50 mcg or 25 mcg of mRNA encoding for the spike protein of SARS-CoV-2 KP.2 variant), Moderna TX Inc.
 - Authorized for use in those 12 years of age and older (50 mcg) and in children 6 months to 11 years of age (25 mcg)
- **Nuvaxovid®** (Novavax) recombinant protein vaccine
 - Authorized for use in those 12 years of age and older
- Side effects – Majority of vaccines share similar side effects but are not limited to:
 - Injection site swelling/pain/redness, hypersensitive reaction, fever, flu-like symptoms, gastrointestinal symptoms, muscle aches, lymph node swelling, loss of appetite

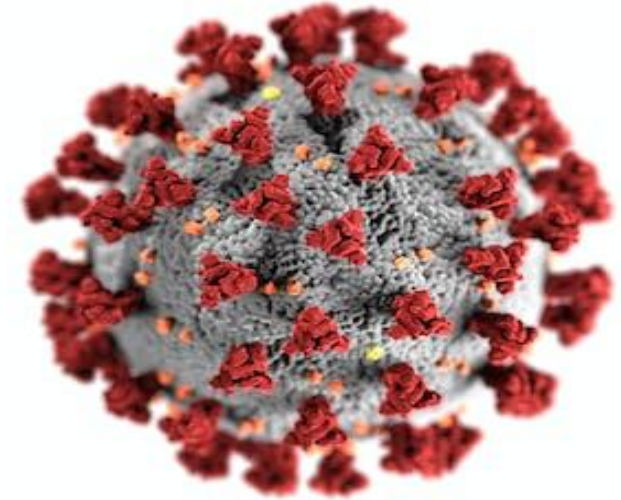
MRNA VACCINE MYTHS

FYI

- 'The vaccine was developed too quickly'
- 'mRNA vaccines change your DNA'
- 'mRNA vaccines have dangerous ingredients'
- 'It's not safe until we get more long-term data'

HERD IMMUNITY

- Indirectly protects populations that may be at a higher risk for severe disease
- Occurs when sufficient percentage of population has become immune to infection
- Can eradicate diseases



FIRST LINE PHARMACOLOGICAL THERAPY-ANTIVIRALS

- Veklury® (**remdesivir**) – injection (IV infusion)
 - AE: Elevated transaminases, headache, nausea, rash
 - **Caution in patients with eGFR <30 mL/min (need to closely monitor)**
 - Chloroquine and hydroxychloroquine may decrease the effect of remdesivir.
May be prone to interactions with CYP450 inducers and inhibitors
- Paxlovid® (**nirmatrelvir & ritonavir**) – tablets (5-day course)
 - Used in non-hospitalized high-risk adults within 5 days of symptom onset
 - AE: Mild sense of altered taste (**metallic taste**), elevated BP, N/V/D, headache, muscle pain, and long-term hepatotoxicity seen with ritonavir
 - Dose reduction in patients with moderate renal impairment (eGFR ≥ 30 to < 60 mL/min)
 - **Not recommended in patients with eGFR <30 mL/min** (although may be used in practice)
 - Both medications are **CYP3A substrates** and may result in decreased plasma levels by CYP3A inducers
 - **Ritonavir is a CYP3A inhibitor** and may increase plasma levels of drugs metabolised by CYP3A

MANAGING DRUG INTERACTIONS WITH PAXLOVID®

- Is it safe to stop the interacting drug for 7 days?
- If you cannot hold the interacting drug: use alternative therapy OR modify therapy by adjusting dose/ using an alternative agent temporarily
- Implement monitoring plan: therapeutic drug monitoring for drugs with narrow therapeutic index, efficacy and safety parameter
- Ontario COVID-19 Science Advisory Table has a helpful tool on managing drug interactions

2ND/3RD LINE PHARMACOLOGICAL THERAPY-IMMUNOSUPPRESSANTS/MABS

- **Dexamethasone** (corticosteroid)
 - Hospitalized patients
 - AE: GI side effects, insomnia, mood disturbances (short-term course side effects)
- **Actemra® (Tocilizumab)** (IL-6 inhibitor)
 - Hospitalized patients who are receiving systemic corticosteroids and require supplemental oxygen
 - AE: Constipation, N/V/D, dysphagia, abdominal pain, UTI, pneumonia, rash, etc.

CHILDREN

- Most children can be managed at home with **non-pharmacological measures** and do not need pharmacological therapy (aside from those used in self-care such as acetaminophen, ibuprofen)
- Those at high risk of severe COVID-19:
 - Age ≤ 1 year of age
 - Obesity
 - On mechanical ventilation
 - Have comorbidities (including but not limited to): cardiac or lung disease, diabetes, CKD, sickle cell anaemia, immunosuppressive therapy, cancer, etc.
 - Genetic, metabolic, neurologic or neuromuscular disorders

PREGNANCY AND BREASTFEEDING

- There is limited evidence available on treatment in pregnant and breastfeeding women
- A **non-pharmacological** approach is first-line:
 - Frequent hand washing
 - Washing hands before handling baby bottles, breast pump or other equipment
 - Wearing a mask while holding or feeding the baby
 - Good breast and skin cleaning practices prior to holding or feeding the baby
 - Ensuring breast pumps, bottles and any other supplies are clean and not shared with others

PREGNANCY AND BREASTFEEDING

Some pharmacological therapy with evidence in this population:

- Dexamethasone
- Veklury® (**remdesivir**)
- Paxlovid® (**nirmatrelvir & ritonavir**)
- Actemra® (Tocilizumab) (IL-6 inhibitor) (unlicensed indication)
- COVID-19 vaccination – Prevention therapy
- Theoretically, due to mAbs having a large size (avg. 150 kDa), they will have a less likely chance to be excreted in breast milk however there is not enough evidence to support this and each drug should be used when the benefit outweighs the risk

MONITORING PARAMETERS

EFFICACY ENDPOINTS		
Parameter	Degree of Change	Timeframe
Cough	Eliminate	Several weeks to months
SpO ₂	Normalize to baseline	Within a few days Full resolution expected in 7-10 days
Temperature (PO)	Normalize to $\leq 37.5^{\circ}\text{C}$	
Other symptoms (sore throat, runny nose, headache, muscle aches)	Improve/eliminate	
Post-acute sequelae (e.g. SOB, fatigue, smell/taste disturbances, cognitive/neurological deficits)	Improve	> 3 months

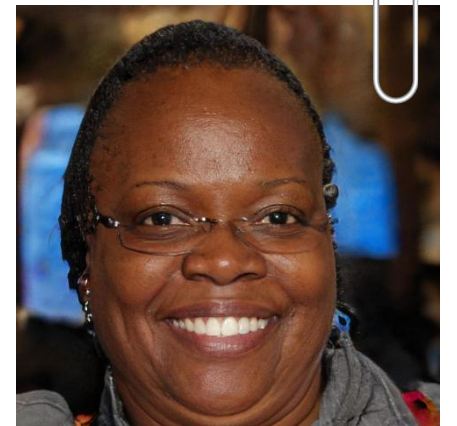
MONITORING PARAMETERS

SAFETY ENDPOINTS		
Parameter	Degree of Change	Timeframe
Adverse effects (headache, diarrhea, nausea/vomiting, rash)	Prevent or minimize	Duration of therapy
Hepatotoxicity	Prevent	
Drug interactions	Prevent	

CASE

RF is a 65-year-old female (BMI=33 kg/m²) with a history of type 2 diabetes, hypertension, rheumatoid arthritis, and dyslipidemia. She had a PCR COVID-19 test today and it was positive. She strongly dislikes hospitals and would like to avoid them if possible. She has a 4-day history of shortness of breath, fever and fatigue. She does not currently require supplemental oxygen. She is also up-to-date with her COVID-19 vaccinations. Her medications include dapagliflozin 10 mg PO daily, amlodipine 10 mg PO daily, methotrexate 15 mg once weekly, adalimumab 40 mg SC every other week, atorvastatin 40 mg PO daily and folic acid 1 mg PO daily except on the day methotrexate is taken.

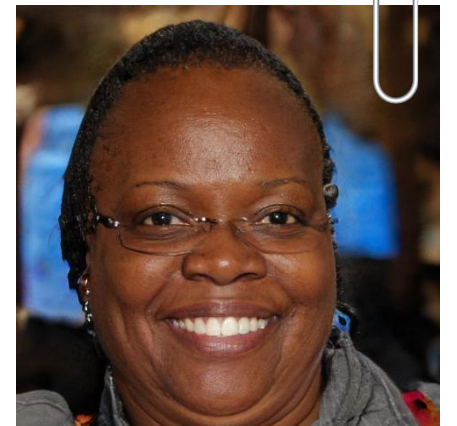
1. Is COVID-19 pharmacological therapy currently indicated for RF?
 - a) Therapy is needed and should be started immediately
 - b) Therapy is needed but can be delayed
 - c) Therapy is not needed, she can be managed with non-pharmacological measures only
 - d) An additional COVID-19 booster vaccine can be given to manage her symptoms



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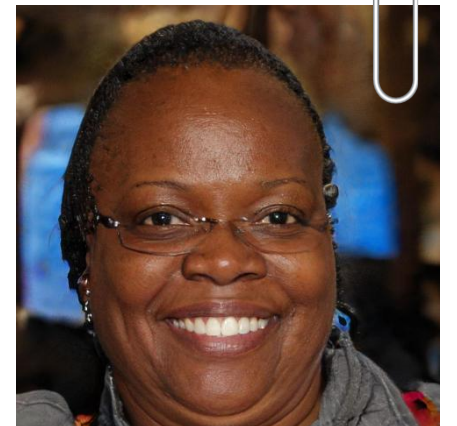
2. What is the best option given the patient's history?
- a) Veklury® (Remdesivir)
 - b) Actemra® (Tocilizumab)
 - c) Paxlovid® (nirmatrelvir/ritonavir)
 - d) Dexamethasone



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3. How can the interaction between the chosen therapy and amlodipine be managed?
- a) Switch amlodipine to ramipril
 - b) Reduce the dose of the amlodipine
 - c) Increase the dose of the amlodipine
 - d) Suggest alternative COVID-19 therapy



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CHANGE LOG

July 2025

- Content reviewed; references updated

September 2025

- Slide 3: Updated legend
- Slides 25-26: Added monitoring parameters slides

February 2026

- Slide 2: Updated NAPRA competencies
- Slides 7, 17: Added FYI
- Slide 19: Updated information on Veklury®. Added 5-day course to Paxlovid® and added dose reduction with moderate renal impairment
- Slide 31: Updated references
- Minor formatting changes made throughout